

Wear and friction of PEEK composites, dry or lubricated

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ABSTRACT

The present study investigates the wear and friction properties of Poly Ether Ether Ketone (PEEK) sliding against stainless steel. The materials are neat PEEK, PEEK with 10% Polytetrafluorethylene (PTFE), PEEK with 30% carbon fibers (CF) and PEEK with 30% glass fibers (GF). Adding fibers to PEEK (CF or GF) increased the wear by a factor of four. This may seem surprising but it is in accordance with the Ratner-Lancaster correlation which postulates that the wear rate is inversely proportional to the product of the strain and strength at break. Adding fibers may increase the strength but it decreases the strain at break even more. Water lubrication increases the wear by a factor of ten, except in the case of CF where the wear is halved. To investigate the influence of the polarity of the lubricant, the very polar water has been replaced by the quasi-nonpolar n-heptane. This decreases the wear significantly. The n-heptane lubricated PEEK with CF shows the lowest wear rate in the present study.

In general the addition of a liquid lubricant decreases the friction whereas it is the opposite when fibers are added. But friction seems to be decoupled from the wear response in the present study.

1. Introduction

Polymers and their composites are used more widely than ever. This is due to the relatively high strength, low density and tribological performance of polymers and their composites compared to monolithic metals [1]. Other characteristics, such as resistance to corrosion and high shock absorbercy, have made polymers popular in the engineering industry [2]. In addition, the material properties of the polymers can be tailor-made to the requirements of the application by incorporating fillers, additives and fibres. In some cases, the polymer has to perform in various liquids. One of these could be in hydropower plants, where bearings could be lubricated and cooled using water. The use of polymer bearings also reduces the severity of an eventual failure, as an oil leak is not at risk of contaminating the water. Hence, a better understanding of the tribological performance of polymers and their composites in lubricated conditions is desired.

In this study, the tribological behaviour of the engineering plastic polyetheretherketone (PEEK) and some of its composites are tested in dry sliding and lubricated conditions. The lubricants used are water and n-heptane. These are selected to test a polar and non-polar liquid as Lancaster [3] did for other materials than PEEK and its composites where some organic liquid and composite combinations showed improved wear properties over water lubrication.

PEEK is a semi-crystalline thermoplastic polymer with a high glass

transition temperature, a high melting temperature, and high mechanical strength [4]. It is particularly resistant to organic and aqueous environments, as well as thermal degradation. PEEK is often utilised in conditions where the chemical resistance combined with the high strength and high operative temperature are needed, which makes it a very interesting material for the engineering industry, although it still is far more expensive than the standard thermoplastic materials such as polyamide (PA), polyoxymethylene (POM) and polyethylene terephthalate (PET) [5].

In terms of wear and friction properties, the sought-after properties of PEEK and its composites generally include low wear and low friction. PEEK can be processed using the widest range of production methods among thermoplastics [6] and can replace metal in some applications [2]. When exposed to sliding wear it is a promising rubbing material [7] but Lancaster [8] reported that fluids like water and other aqueous solution inhibit the formation of transfer films on the counterface. Consequently, the wear rates are generally greater in lubricated than in dry condition. Additionally, Tanaka [9] reports that the introduction of water generally reduces the coefficient of friction, but may increase the wear rate of the polymer in a metal-polymer sliding combination. When reinforcing polymers with fibres, not only are the mechanical properties improved, but the wear and friction properties could be as well [10]. The addition of glass fibre to polymers increases the stiffness and tensile strength [11] as well as the glass transition temperature and mechanical

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stability at elevated temperatures [12]. As a composite material, PEEK provides the widest range of mechanical properties achieved by a thermoplastic so far [6]. In dry sliding, Hanchi and Eiss [13] found that adding carbon fibre to the polymer improves the wear resistance and reduces the friction coefficient significantly. Furthermore, Lu and Friedrich [14] found that the addition of PTFE to PEEK reduced the wear rate volume fraction with up to 85%, compared to neat PEEK, when sliding against a hard steel counterface. The optimal amount of PTFE added to PEEK for lowering the wear rate and friction, ranges between 10 wt% and 30 wt% as shown by Burris [15] and Stuart [16]. Additionally, Burris and Stuart found that the friction coefficient of a PEEK-PTFE composite is lower than either of the materials when neat. The reduction in wear and friction for the different composites is considered to be caused by a lubricating transfer film on the steel counterface, for both PTFE and fibre filled PEEK [16]. Like neat PEEK, its composites show an increased wear rate when water is introduced [17]. Lancaster [8] points out that water inhibit the formation of transfer film of the filler/polymer debris on the counterface. In addition, Davim, Marques, and Baptista [18] observed cavities around the fibre boundary on the rubbing surface of carbon fibre reinforced PEEK sliding on steel in water. Thus, a separation between filler and polymer is thought to contribute to the increased wear, when in water. Yamamoto and Hashimoto [10] point out that the use of distilled water can reduce the friction of filled PEEK but at the expense of a large reduction in wear resistance. It seems that the benefits of the reinforcement of polymers depend on the operating conditions, but the wear properties are generally worse in water lubrication [10].

When performing the experiments to compare the sliding friction and wear of the polymers, it is important to maintain the same lubrication condition. Furthermore, it is important to check that boundary lubrication occurs when adding a lubricant, like seen in the experiments of Yamamoto and Hashimoto [4,10]. Hydrodynamic lubrication must be avoided, as it would be an indication of the lubricant viscosity, not of the wear extent. To ensure boundary lubrication, the film-thickness of the lubricant in the sliding contact region has to be thinner than the surface roughness of the two contact faces. Other essential parameters are the sliding velocity and the contact pressure. If the sliding velocity or the contact pressure is too high, the temperature will increase and the polymer will melt on the surface, making the results unusable. This introduces the pv-level as a parameter, which is the product of the contact pressure, p , and the sliding velocity, v .

This is to be kept below a certain value in order to avoid melting the polymer. The value depends on the materials [5]. If the pv-level is too high, a significant improvement in the wear properties when the polymer is lubricated is seen [19]. Though this improvement is not in direct relation to the material properties, but closer related to the reduction of the temperature at which the polymer operates. Furthermore it is observed by Lu and Friedrich [14] that the wear rate increases with the pv-level, but not the friction. The pv level used in this study is in the range of 1–1.3. This is significantly lower than run by Wang et al. [20, 21] and similar to the range run by Yamamoto and Hashimoto [4]. In the present study, the variables are kept constant by only changing one parameter at the time. Therefore, the composites are selected to include only one filler e.g. carbon fibre or glass fibre and not a combination of these. This includes the material sliding on the polymers as well, as it can be hard to compare the results for e.g. steel with stainless steel [17]. Furthermore, the surface roughness has to be kept constant, as the wear rate is highly dependent on the surface roughness, as observed by Ovaert and Cheng [22]. A way to keep the surface roughness constant is to machine the sliding surface on the contact wheel in the same way every time. Moreover, machining between every experiment helps removing the excess polymer from previous experiments. Another variable to bear in mind is the fibre orientation in the fibre reinforced polymer. Nak-Ho and Suh [23] found the friction coefficient and wear to be at a minimum when the orientation of the fibres was normal to the sliding surface. The lowest wear resistance occurs when the fibres are oriented in the plane

of the sliding contact and antiparallel to the sliding direction [24]. The effect of fiber orientation has not been investigated in the present study.

2. Materials and experimental setup

Four PEEK variants, purchased from Ensinger, have been tested against stainless steel AISI316L. The polymers are PEEK, PEEK with 10% PTFE (polytetrafluorethylene) (PEEK PTFE10), PEEK with 30% carbon fibres (PEEK CF30) and PEEK with 30% glass fibres (PEEK GF30). Some of their properties are shown in Table 1.

The properties in Table 1 have been gathered from data sheets from Ensinger.

The principle of the experimental setup is shown in Fig. 1.

The 8 mm diameter polymer pin is sled against a rotating contact wheel (cylinder) made of stainless steel.

The actual setup is shown in Fig. 2.

Fig. 2 shows the contact wheel which slides against the polymer pin. The contact is hidden by a container in which a lubricant can be added. A normal force load cell is fixed to the pneumatic piston which delivers the load on the contact. The other load cell gives the friction force. The setup originates from an earlier study [25].

The sliding velocity is 30 mm/s and the load is 200 N. The pressure velocity PV value is 1 to 1,3. According to engineering brochures this value is acceptable [5] which was also mentioned in the introduction. The duration is 2,5 h which gives a sliding distance of 270 m. The temperature is ambient meaning $23\text{C} \pm 3\text{C}$. The maximum Hertzian pressures ranges from 35 MPa (PEEK PTFE10) to 46 MPa (PEEK GF30) [26]. These pressures are way below the yield stresses indicated in table and they will drop drastically with the onset of wear. The wear extend is evaluated by measuring the width of the wear scar through a microscope as it is shown in Fig. 3 for a test with PEEK.

The principle in the establishing of the wear volume is shown in Fig. 4.

By measuring the wear scar width W , the angle θ can be calculated:

$$\theta = 2 \arcsin\left(\frac{W}{2r}\right) [\text{rad}]$$

The wear section A is then given by:

$$A = \frac{1}{2} r^2 (\theta - \sin(\theta)) [\text{mm}^2]$$

The wear volume V is then

$$V = AL [\text{mm}^3]$$

Where L is the diameter of the pin in mm. The above is an approximate calculation as it is valid for a pin with a square section. Accurate calculation for a circular pin (not shown here) gives an error of less than five percent at a wear scar width of 5000 μm . Therefore the approximate solution is used in the present work.

After each test the cylinder is machined to cut off 0,1 mm of metal. This is to ensure that no polymer is left on the surfaces. The rotational speed of the wheel is adjusted to keep the same linear speed. The surface of the steel is quite smooth as a R_a value of 0,287 μm has been measured. The roughness of the pin has not been established but the pins are all machined in the same way. The coefficient of friction (CoF) is measured during the test by dividing the friction force by the load. Fig. 5 shows the normal load and the CoF as a function of time for a test with PEEK.

Table 1
The tested PEEK variants.

PEEK with	E modulus (MPa)	Stress at yield (MPa)	Strain at break (%)
–	4200	116	15
10% PTFE	3600	96	7
30% CF	6000	112	10
30% GF	6300	113	5

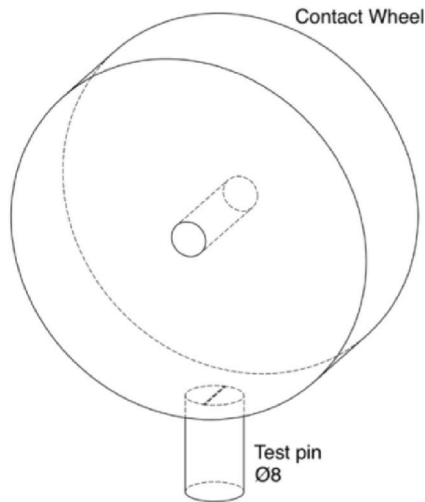


Fig. 1. The pin on cylinder test setup.

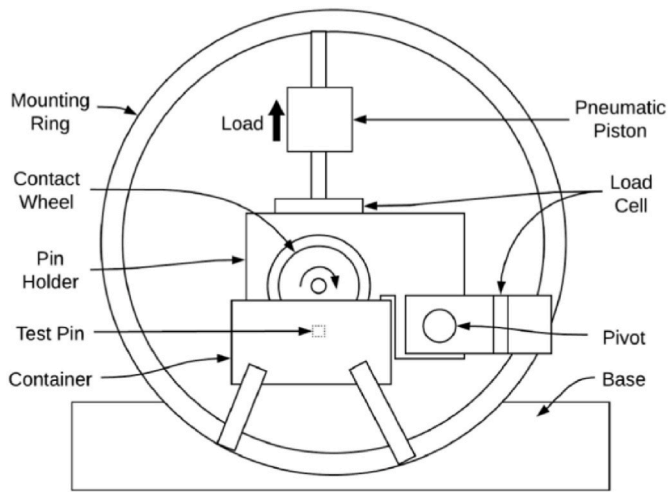


Fig. 2. The complete experimental setup.

3. Results

The specific wear of the polymers is shown in Fig. 6.

The specific wear rate in Fig. 6 is defined as:

$$w = \frac{V}{Fd} \left[\frac{\text{mm}^3}{\text{Nm}} \right]$$

Where V is the wear volume in mm^3 , F the applied load in N and d the sliding distance in m . The variance in the measurements is expressed as the covariance CoV :

$$\text{CoV} = \frac{s}{\bar{x}} [-]$$

Where s is the standard deviation and $\bar{x}$ the average of the wear scar width, both in μm . For four consecutive tests with PEEK (with renewed wheel and pin each time) the covariance has been calculated to be 0,065 which corresponds to 6,5%. The assumption that the covariance has the same deviation in percent for all the measurements has resulted in the error bars in Fig. 6. The maximum wear rate is indicated by the height of the column and the minimum is shown as a line.

The indicated coefficient of friction in Fig. 7 is the average over the whole test.

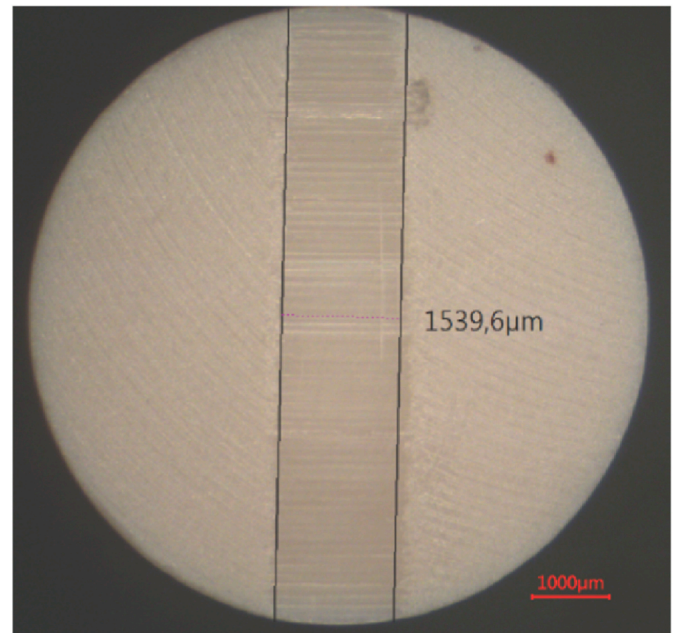


Fig. 3. The wear scar on the pin after a test with neat PEEK.

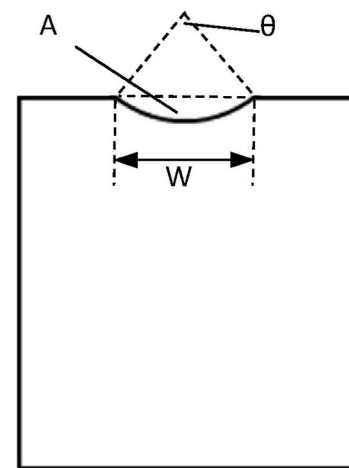
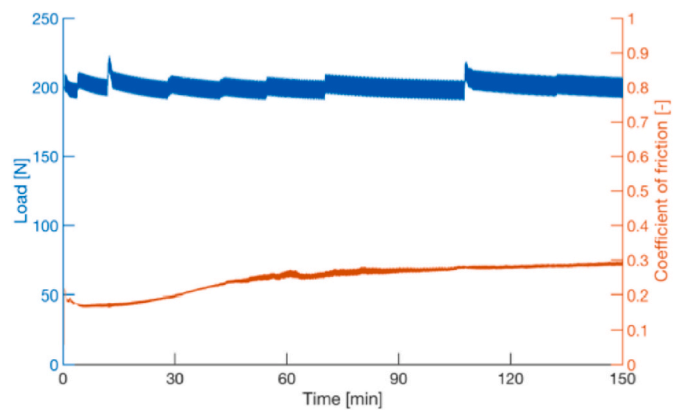
Fig. 4. The pin is shown from the side. From the wear scar width W the angle θ and the wear section can be deduced (see text).

Fig. 5. The normal force (blue, top) and the CoF (red, bottom) as a function of time for a test with neat PEEK. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

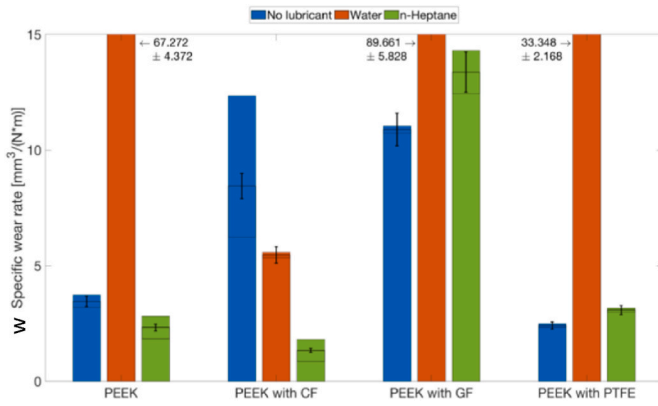


Fig. 6. Specific wear of the polymers dry and lubricated with water and n-heptane respectively.

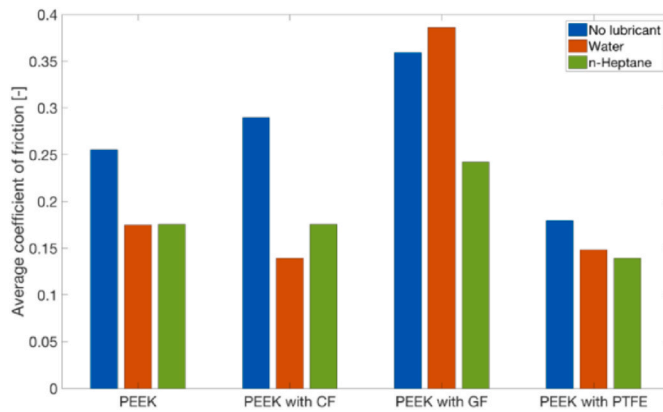


Fig. 7. The CoFs measured during the tests. Under dry conditions and lubricated with water and n-heptane respectively.

4. Discussion

It can be noticed in Table 1 that all the polymers have the same strength at yield except for PEEK PTFE, which has a lower one. The fibres are added to increase the E-modulus which means that, in the case of carbon, the fibres are of type I (high-heat-treatment carbon fibers). If the strength would have been higher the strain at break would have been 1–2%. For the here studied polymers with fibers, it is 5–10%.

Lancaster [27] showed in the 1960s, very much inspired by Ratner [28], that the abrasive wear rate of 18 tested polymers was a function of the reciprocal of the product of the strength σ_B [MPa] and strain at break ϵ_B [–], here called the Ratner-Lancaster relation (RL).

$$RL = \frac{1}{\sigma_B \epsilon_B} [MPa^{-1}]$$

This equation is used for calculating the RL of the three of the studied polymers. For neat PEEK the RL is 0,06, for PEEK CF30 0,09 and for PEEK GF30 0,18, all in MPa^{-1} . The trend in the dry wear rates in Fig. 6 also indicates that PEEKs with fibers wear more. For PEEK PTFE10 it is known that PTFE is transferred to the metallic surface so the sliding is done in a PTFE interface [29]. The RL cannot predict this which is why Lancaster only performed single pass experiments. It should also be noticed that PEEK PTFE10 has the highest wear resistance when it comes to dry sliding.

Fig. 6 shows that the specific wear increases with a factor of 8 or more when water lubrication is introduced. Only the PEEK CF30 show a decreasing wear rate. Lancaster [8] argued that water inhibits the formation of a transfer film but the mechanism is not known. It can be seen

that water decreases the wear resistance more in PEEK contacts than it is the case for PEEK GF30 and PEEK PTFE. In the former case the increase is more than a factor of 19 whereas it is 8 for PEEK GF30 and 14 for PEEK PTFE10. PEEK CF30 seems to form a different wear resistance system as the addition of water decreases the wear rate compared to dry sliding. This latter result gives hope for water lubricated bearings in hydropower applications. Another study has also found that water lubrication decreases the wear rate for PEEK with carbon fibres compared with neat PEEK [10]. But, in general, adding fibres to polymers increases the wear rates in water [8]. This difference seems to depend on the studied polymers and the test conditions so no adequate explanation can be given here.

If n-heptane is used as lubricant the wear decreases for PEEK and PEEK CF30 compared to dry sliding. For PEEK GF30 and PEEK PTFE10 it is the opposite. The surface energy of n-heptane is more than three times lower than the one of water: 20 mN/m against 72,8 mN/m [30,31]. So n-heptane will wet the surfaces better than water. There is no proven relation between wetting and wear but a previous study indicates that the wear of polymers is decreased when the surface energy of the lubricant is decreased [3].

Fig. 7 shows that adding fibres to PEEK increases the friction in dry sliding. So the friction of the fibres against the metal is in both cases higher than for the polymer. Adding PTFE decreases the friction compared to the PEEK but when lubricated this improvement is very small.

The viscosity of n-heptane is less than half of that of water: 0,389 mPas compared to 0,89 mPas [32,33]. Fig. 6 indicates that hydrodynamic lubrication isn't predominant as the lower viscosity n-heptane results in lower wear rates than water. Fig. 6 also shows this as the friction is lower for the n-heptane lubricated systems in most of the cases. Calculations of the film thicknesses in the contacts [34] confirms this as these do not exceed one nm in any of the measurements. The calculation of the minimum film thickness for neat PEEK in water is shown here. All equations are taken from Ref. [34]. The effective elastic modulus is given by

$$E' = \frac{2}{\frac{(1-\nu_1^2)}{E_1} + \frac{(1-\nu_2^2)}{E_2}}$$

Where E_1 and ν_1 are the elastic modulus and poisson ratio of the steel equal to 193 GPa and 0,27. The values for PEEK indexed 2 are 4,2 GPa and 0,39. This gives an E' of 9,68 GPa. The effective radius in the rolling direction is r the radius of the wheel. The dimensionless minimum film thickness H is given by:

$$H = 1,714 W'^{-0,128} U^{0,694} G^{0,568}$$

Where W' is the load parameter:

$$W' = \frac{P'}{E r}$$

P' is the load per unit length or $200/0,008 = 25000$ N/m. W' is then $1,005E-4$.

U is the speed parameter:

$$U = \frac{\eta_0 u}{E r}$$

η_0 is the viscosity at ambient pressure and u is the rolling speed which is half of the sliding speed. So U becomes $5,37E-14$.

G is the material parameter:

$$G = \xi E'$$

Where ξ is the pressure viscosity coefficient here estimated at $1,6$ GPa^{-1} (one tenth of that of n-heptane). So G is 15,48.

The dimensionless film thickness is then $1,63E-8$. The minimum film thickness h is then:

$$h = Hr$$

Or 0,42 nm. The film thickness is about two times larger when n-heptane is used as lubricant due to the larger pressure viscosity coefficient. In any case the minimum film thickness is hundreds times smaller than the roughness of the surfaces so hydrodynamic lubrication isn't prevalent.

5. Conclusions

In this study, a pin-on-cylinder apparatus was used to examine the wear friction properties of PEEK and some of its composites in dry and lubricated conditions. In terms of wear and friction properties, the sought-after properties of PEEK and its composites generally include low wear and low friction. Hence, a single material cannot be said to be superior in all lubricating conditions in this regard. When the sliding is not lubricated, PEEK PTFE10 is superior in terms of wear and friction. When lubricated with water, PEEK CF30 is superior. Moreover, PEEK CF30 is the best choice in n-heptane lubrication. The wear is lower than in water lubrication at the expense of higher friction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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